

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1  
getwd()
```

```
## [1] "/home/guest/ENV872/EDE_Bella"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.1      v tibble     3.2.1  
## v lubridate  1.9.3      v tidyr      1.3.1  
## v purrr      1.0.2  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
library(here)
```

```
## here() starts at /home/guest/ENV872/EDE_Bella
```

```
library(ggplot2)

lake_chem_phy <- read.csv(here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"))

#2
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        panel.background = element_rect(fill = "gray95", color = NA),
        plot.background = element_rect(fill = "white", color = NA),
        plot.title = element_text(
          hjust = 0.5,
          face = "bold"),
        legend.position = "right")
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: $H_0: \beta_1 = 0$ $H_a: \beta_1 \neq 0$
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
july_lake <- lake_chem_phy %>%
  mutate(sampledate = as.Date(sampledate, format = "%m/%d/%Y")) %>%
  filter(format(sampledate, "%m") == "07") %>%
  select(lakename, year4, daynum, depth, temperature_C) %>%
  drop_na()

#5
depth_temp <-
  ggplot(july_lake,
        aes(x=depth, y=temperature_C))+
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  geom_point(alpha = 0.5, color = "steelblue", size = 1.5)+
  ylim(0, 35)+
  mytheme+
```

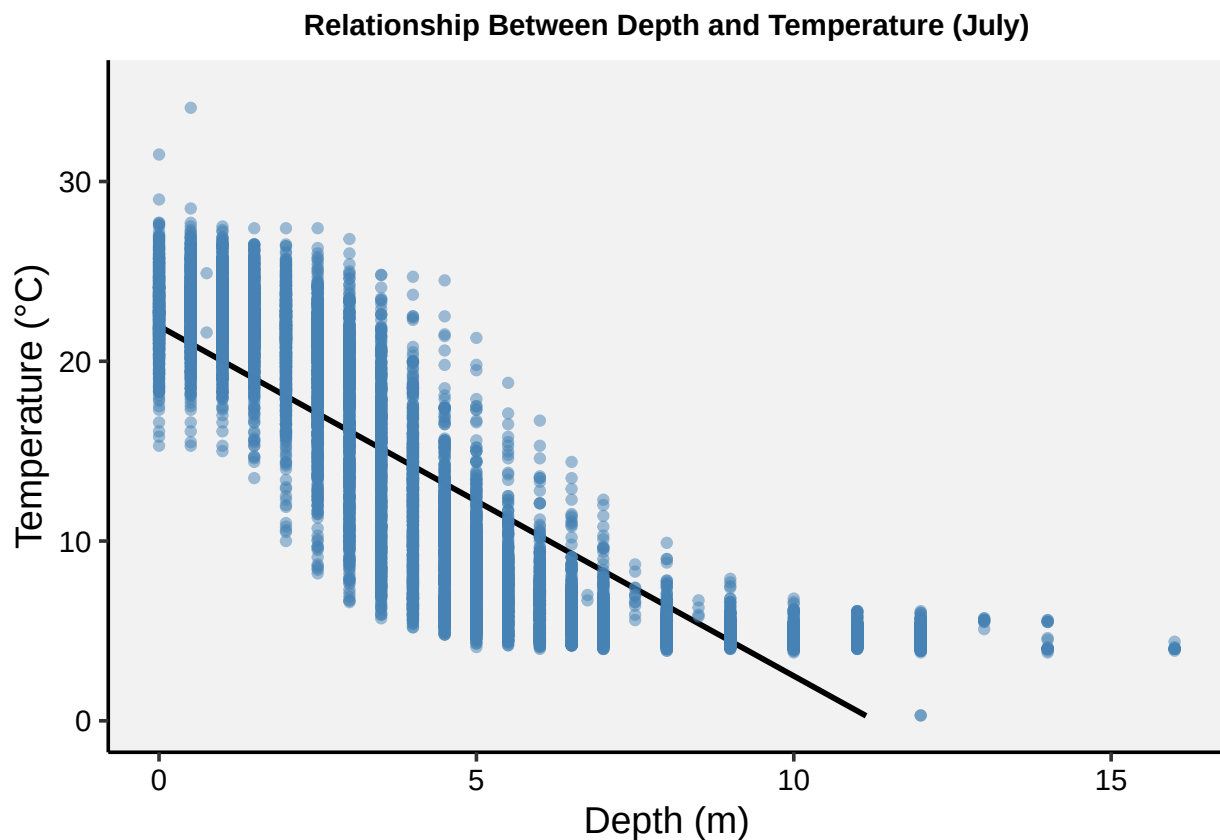
```

labs(
  title = "Relationship Between Depth and Temperature (July)",
  x = "Depth (m)",
  y = "Temperature (°C)" ) +
  theme(
    plot.title = element_text(size=11)
  )
print(depth_temp)

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: When depth increases, the temperature decreases in July. The regression line we plot has a negative slope, and this correlation is relatively strong. However, from the distribution of points, I can tell that it is not perfectly linear, since in shallower water, where the water depth is small, the temperature seems to decrease at a smaller rate than at a deeper depth.

7. Perform a linear regression to test the relationship and display the results.

```

#7
depth_temp_regression <- lm(
  data = july_lake,
  temperature_C ~ depth)
summary(depth_temp_regression)

##
## Call:
## lm(formula = temperature_C ~ depth, data = july_lake)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16

```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: the coefficient of the regression is -1.95, and the intercept is 21.96, which means that for every meter increases in depth, the temperature would decrease by 1.95 degree roughly. According to the summary of the model, the p-value for the regression is $< 2.2e-16$, which means that the negative correlation between the depth and the temperature is highly statistically significant. The adjusted R-squared is 0.7387, so 73.87% of the variability in lake temperature is explained by changes in depth. The degree of freedom for this model is 9726, resulting from the total 9728 observation and two variables.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

#9

```
tempByAll_reg <- lm(data=july_lake,  
                    temperature_C ~ depth + year4 + daynum)
```

```
step(tempByAll_reg)
```

```
## Start: AIC=26065.53  
## temperature_C ~ depth + year4 + daynum  
##  
##           Df Sum of Sq    RSS   AIC  
## <none>                 141687 26066  
## - year4    1         101 141788 26070  
## - daynum   1         1237 142924 26148  
## - depth    1      404475 546161 39189  
  
##  
## Call:  
## lm(formula = temperature_C ~ depth + year4 + daynum, data = july_lake)  
##  
## Coefficients:  
## (Intercept)      depth      year4      daynum  
##   -8.57556    -1.94644     0.01134     0.03978
```

The the AIC for the model that includes all the variables is 26065.53, lower than the other AICs. Thu

#10

```
temp_multireg <- lm(data=july_lake,  
                   temperature_C ~ depth + year4 + daynum)
```

```
summary(temp_multireg)
```

```
##  
## Call:  
## lm(formula = temperature_C ~ depth + year4 + daynum, data = july_lake)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -9.6536 -3.0000  0.0902  2.9658 13.6123   
##  
## Coefficients:  
##              Estimate Std. Error  t value Pr(>|t|)      
## (Intercept) -8.575564   8.630715  -0.994   0.32044      
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***  
## year4        0.011345   0.004299   2.639   0.00833 **   
## daynum       0.039780   0.004317   9.215   < 2e-16 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 3.817 on 9724 degrees of freedom  
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411   
## F-statistic: 9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The the AIC for the model that includes all the variables is 26065.53, lower than the other AICs. Thus, all of the three variables should be included in the regression for a better interpretation. Now the adjusted r-squared is 0.7411, representing that 74.11% of the observed variance is explained by this new multi-variable regression. This is an improvement, even though a small one, and the depth remain as the dominant explanatory variable.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

```
#12
# Format ANOVA as aov
july_lake_anova <- aov(data = july_lake, temperature_C ~ lakename)
summary(july_lake_anova)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Format ANOVA as lm
july_lake_anova2 <- lm(data = july_lake, temperature_C ~ lakename)
summary(july_lake_anova2)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = july_lake)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake  -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake     -6.5972     0.6769  -9.746 < 2e-16 ***
```

```
## lakenameWard Lake      -3.2078      0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake -6.0878      0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: According to the result of the Anova2, the p-value < 2.2e-16, so it is statistically significant that there is a difference in mean July lake temperature among lakes. In other word, at least one lake has a statistically significant different mean temperature from the others. However, the adjusted r-squared is 0.03874, which means that the lake identity only represents 3.87% of the total variation in temperature.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
lake_depth <-
  ggplot(july_lake, aes(x=depth, y=temperature_C, color=lakename))+
  geom_point(alpha = 0.5, size = 2)+
  geom_smooth(method = "lm", se = FALSE)+
  ylim(1,35) +
  labs(
    title = "Relationship Between Depth and Temperature (July)",
    x = "Depth (m)",
    y = "Temperature (°C)",
    color = "Lake Name" )+
  mytheme +
  theme(plot.title = element_text(size=11))

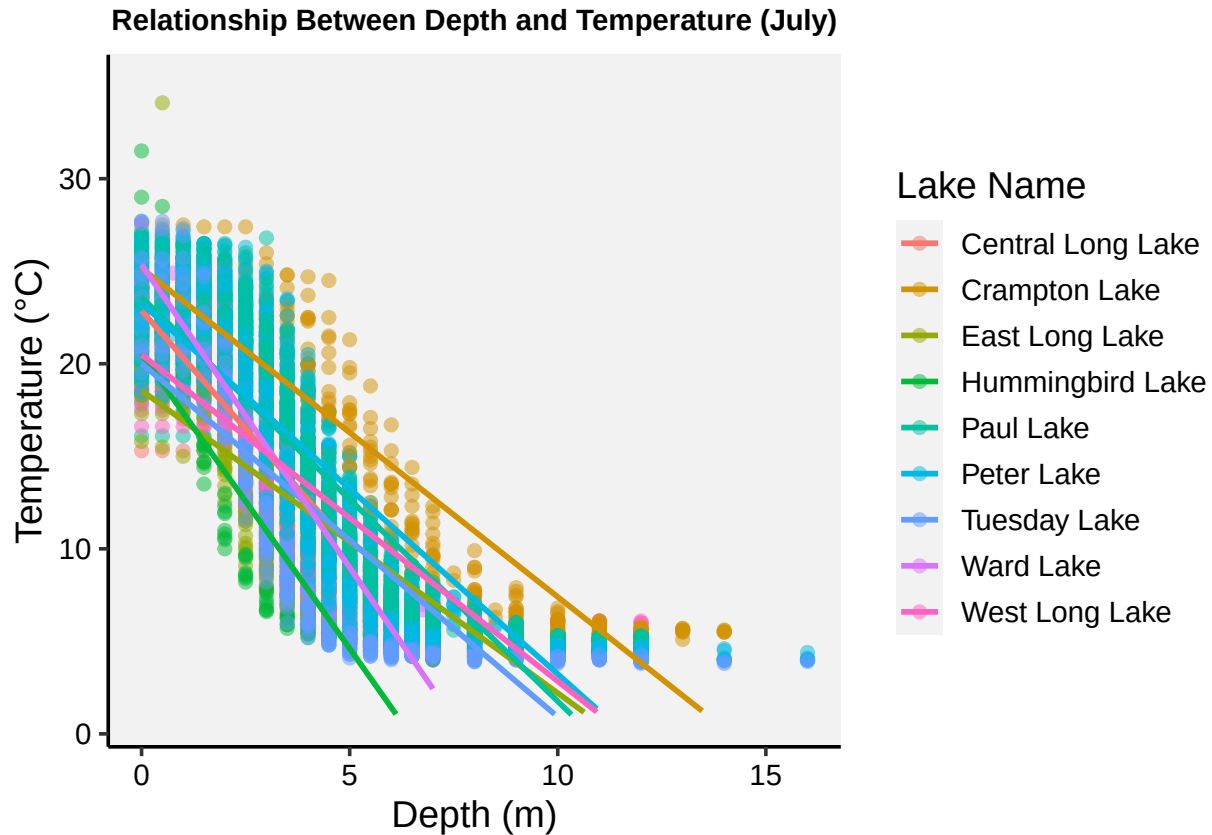
print(lake_depth)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## ('geom_point()').
```

```
## Warning: Removed 95 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

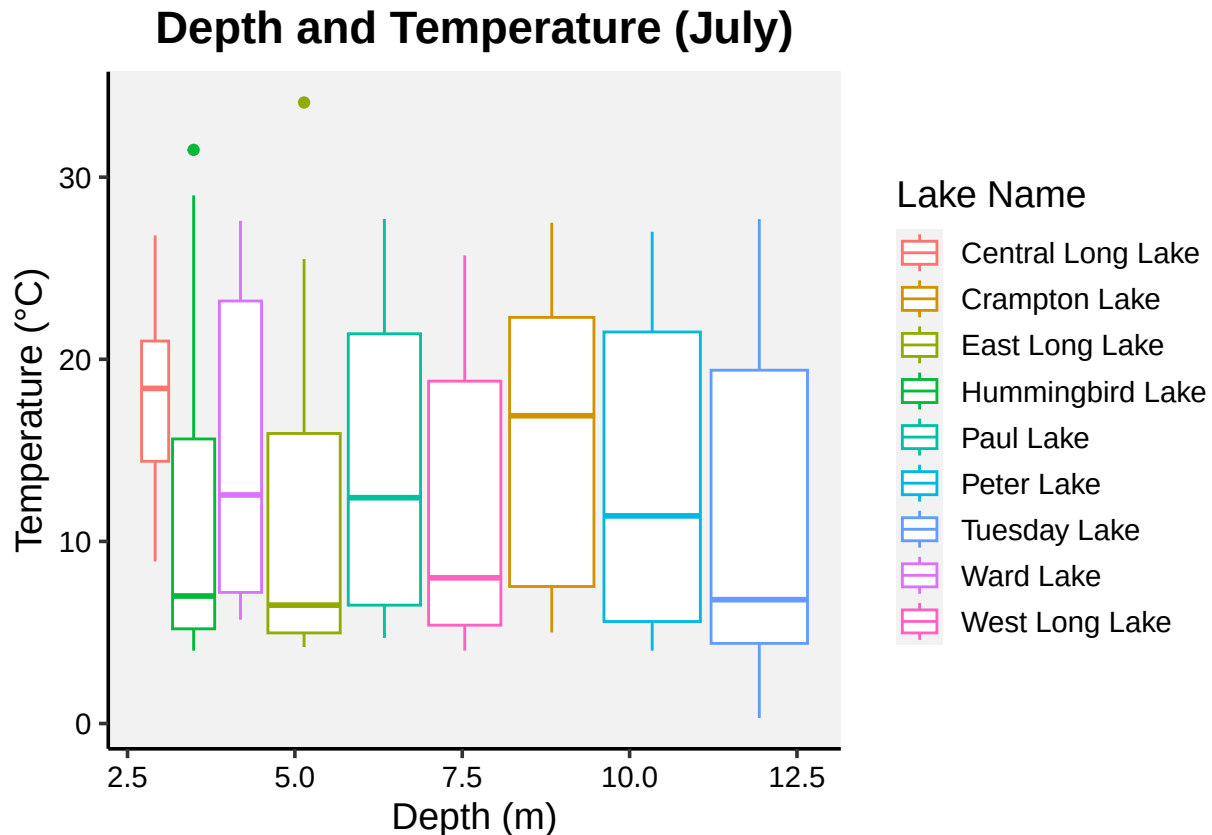
```
#15
library(agricolae)
TukeyHSD(july_lake_anova)

##      Tukey multiple comparisons of means
##      95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = july_lake)
##
## $lakename
##              diff          lwr          upr      p adj
## Crampton Lake-Central Long Lake -2.3145195 -4.7031913  0.0741524 0.0661566
## East Long Lake-Central Long Lake -7.3987410 -9.5449411 -5.2525408 0.0000000
## Hummingbird Lake-Central Long Lake -6.8931304 -9.8184178 -3.9678430 0.0000000
## Paul Lake-Central Long Lake      -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake      -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake    -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake       -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake  -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake      -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake    -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake           -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake          -2.0356263 -3.3842699 -0.6869828 0.0000999
```


## Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000
## Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
## West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
## Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000
## Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
## Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
## Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
## West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

```
july_anova_plot <-
  ggplot(july_lake, aes(x=depth, y=temperature_C, color = lakename)) +
  geom_boxplot() +
  mytheme +
  labs(
    title = "Depth and Temperature (July)",
    x = "Depth (m)",
    y = "Temperature (°C)",
    color = "Lake Name" )

print(july_anova_plot)
```



16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: In order for two lakes to have a statistically same mean, the p-adj for the comparison should be greater than 0.05 to indicate no significant difference. For Peter Lake, among all the other lakes, the Paul Lake (p adj=0.224) and Ward Lake (p adj=0.783) have similar mean July temperatures. While there are some lakes that show significant difference in average temperature, none of the lakes have a mean July temperature that is completely statistically distinct from all other lakes.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: If we are just looking at the two lakes, we can use two-sample t-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match your answer for part 16?

```
Crampton_ward <- july_lake %>%
  filter(lakename == "Crampton Lake" | lakename == "Ward Lake")

CW_twosample <- t.test(data=Crampton_ward, temperature_C ~ lakename)
CW_twosample
```

```
##
## Welch Two Sample t-test
##
## data: temperature_C by lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.35189                14.45862
```

Answer: With a p-value of 0.2649, we failed to reject the null hypothesis that there is no statistically significant difference in mean July temperature between Crampton Lake and Ward Lake. In the 95% confidence interval, it includes 0, so the mean temperatures for the lakes are statistically equal. It matches my result in part 16, where the p adj for the two lakes is 0.97, indicating no significant difference.